OpenCPU SDK

SIM API Notes

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Notice

This document is specifically for **N706B** and **N520** modules.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for system engineers (SEs), development engineers, and test engineers.

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Wang Chen

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the SIM card operation-related APIs in OpenCPU SDK, covering the following aspects:

- Retrieving SIM card-related information such as IMSI, ICCID, and MSISDN.
- Obtaining SIM card status.
- Retrieving and configuring SIM card PIN status.
- SIM card power on/off operations.
- SIM card switching operations.
- Registering and unregistering for SIM card status notifications.
- APDU pass-through operations for SIM cards

2 Data Structure

2.1 typedef enum

```
typedef enum
{
    NWY_SIM_NOT_INSERTED,
    NWY_SIM_READY,
    NWY_SIM_PIN_REQ,
    NWY_SIM_PUK_REQ,
    NWY_SIM_BUSY,
    NWY_SIM_BLOCKED,
    NWY_SIM_UNKNOWN
}nwy_sim_status_e;

typedef enum
{
    NWY_SIM_PIN_MODE_ENABLED,
    NWY_SIM_PIN_MODE_DISABLED,
    NWY_SIM_PIN_MODE_UNKOWN
}nwy_sim_pin_mode_e;

typedef enum
{
    NWY_SIM_ID_SLOT_1,
    NWY_SIM_ID_SLOT_2,
    NWY_SIM_ID_ESIM,
    NWY_SIM_ID_REMOTE,
    NWY_SIM_ID_MAX
}nwy_sim_id_e;
```

2.2 typedef struct

```
typedef struct {
    char mcc[4];
    char mnc[4];
    int  mnc_len;
}nwy_sim_plmn_t;
```

3 APIs

This interface is defined in nwy_sim_api.h and enables querying and managing SIM/USIM/RUIM cards, including retrieving card status, performing PIN code operations, and reading essential files from the UICC.

3.1 nwy_sim_imsi_get

Format	nwy_error_e nwy_sim_imsi_get(nwy_sim_id_e sim_id, char* imsi_buf, uint8_t imsi_len);
Description	Retrieves the International Mobile Subscriber Identity (IMSI) of the device.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID. char* imsi_buf: Buffer for the IMSI value. uint8_t imsi_len: Length of the IMSI buffer.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.2 nwy_sim_iccid_get

Format	nwy_error_e nwy_sim_iccid_get(nwy_sim_id_e sim_id, char* iccid_buf, uint8_t iccid_len);
Description	Retrieves the Integrated Circuit Card Identifier (ICCID).
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. char* iccid_buf: Buffer for the ICCID value. char* iccid_buf: Buffer for the ICCID value.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.3 nwy_sim_status_get

Format	nwy_error_e nwy_sim_status_get(nwy_sim_id_e sim_id, nwy_sim_status_e *status);

Description	Retrieves the status of the SIM card.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sim_status_e *status: Pointer to store the SIM card status value.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.4 nwy_sim_pin_verify

Format	nwy_error_e nwy_sim_pin_verify(nwy_sim_id_e sim_id, char* pin)
Description	Verifies the PIN code when the UICC state is NWY_SIM_PIN_REQ.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. char* pin: PIN code string.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.5 nwy_sim_pin_unlock

Format	nwy_error_e nwy_sim_pin_unlock(nwy_sim_id_e sim_id, char* puk, char* new_pin)
Description	Unlocks the PIN when the state is NWY_SIM_PUK_REQ using a PUK code.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. char* puk: PUK code string. char* new_pin: New PIN code string to set after unlocking.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.6 nwy_sim_pin_mode_get

Format	nwy_error_e nwy_sim_pin_mode_get(nwy_sim_id_e sim_id, nwy_sim_pin_mode_e *pin_mode)
Description	Retrieves the current PIN code enable state.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sim_pin_mode_e *pin_mode: enable status of the pin code
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.7 nwy_sim_pin_enable

Format	nwy_error_e nwy_sim_pin_enable(nwy_sim_id_e sim_id, char* pin)
Description	Enables the PIN code when the state is NWY_SIM_PIN_MODE_DISABLED.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. char* pin: PIN code string.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.8 nwy_sim_pin_disable

Format	nwy_error_e nwy_sim_pin_disable(nwy_sim_id_e sim_id, char* pin)
Description	Disables the PIN code when the state is NWY_SIM_PIN_MODE_ENABLED.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. char* pin: PIN code string.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.9 nwy_sim_pin_change

Format	nwy_error_e nwy_sim_pin_change(nwy_sim_id_e sim_id, char* old_pin, char* new_pin)
Description	Changes the PIN code. The PIN must be enabled to perform this operation.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. char* old_pin: Current PIN code string. char* new_pin: New PIN code string.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.10 nwy_sim_pin_times_get

Format	nwy_error_e nwy_sim_pin_times_get(nwy_sim_id_e sim_id, uint8_t *pin_times, uint8_t *puk_times)
Description	Retrieves the remaining attempt counts for PIN and PUK.

Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. uint8_t *pin_times: Returns the remaining PIN attempts. uint8_t *puk_times: Returns the remaining PUK attempts.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.11 nwy_sim_msisdn_get

Format	nwy_error_e nwy_sim_msisdn_get(nwy_sim_id_e sim_id, char* msisdn_buf, uint8_t msisdn_len)
Description	Retrieves the MSISDN of the SIM card.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. char * MSISDN_uf: Buffer to store the MSISDN value. uint8_t msisdn_len: Reserved, not in use currently.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.12 nwy_sim_power_set

Format	nwy_error_e nwy_sim_power_set(nwy_sim_id_e sim_id, int power_set)
Description	Controls the power state of the SIM card.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. int power_set: Specifies the power state (0 for power off, 1 for power on).
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.13 nwy_sim_slot_set

Format	nwy_error_e nwy_sim_slot_set(nwy_sim_id_e sim_id, nwy_sim_cfg_type_e save)
Description	Switches the SIM card slot to be used.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to switch to. nwy_sim_cfg_type_e save: Determines whether the slot setting persists after a power cycle.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.14 nwy_sim_slot_get

Format	nwy_error_e nwy_sim_slot_get(nwy_sim_id_e *sim_id)
Description	Queries the currently active SIM card slot.
Parameters	nwy_sim_id_e sim_id: Pointer to store the active SIM card ID.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.15 nwy_sim_csim

Format	nwy_error_e nwy_sim_csim(nwy_sim_id_e sim_id, char *indata, int indata_len, char *outdata, int outdata_len)
Description	Sends data to the SIM card.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.16 nwy_sim_urc_reg

Format	nwy_error_e nwy_sim_urc_reg(nwy_sim_id_e sim_id, nwy_sim_evt_handler sim_reg_func)
Description	Registers a callback function for SIM card events.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sim_evt_handler sim_reg_func: The callback function for event handling.
Return Value	Success: NWY_SUCCESS Failure: another enumeration value

3.17 nwy_sim_urc_unreg

Format	nwy_error_e nwy_sim_urc_unreg(nwy_sim_id_e sim_id, nwy_sim_evt_handler sim_reg_func)
Description	Unregisters a callback function for SIM card events.
Parameters	nwy_sim_id_e sim_id: Specifies the SIM card ID to operate. nwy_sim_evt_handler sim_reg_func: The callback function for event handling.

Return Value	Success: NWY_SUCCESS Failure: another enumeration value

